

Advanced NLP – Exercise 2

Part 2: Sentiment Analysis Failure Analysis

Code: <https://github.com/netanelazran11/ANLP-2026/tree/main/ex2>

The failure type investigated is **implicit negative sentiment**: sentences that are negative in meaning but contain no explicit negative vocabulary. Negativity must be inferred from understatement or real-world consequence (e.g., “*The battery lasts just long enough to get from my bedroom to the kitchen.*”). The hypothesis is that models rely on surface lexical cues; when those cues are absent they default to **Positive**.

Break Examples

Nine sentences with negative meaning but no explicit negative words. Both models exceed the five-failure threshold: Qwen3-8B fails on **8/9**, Llama-3.1-8B-Instruct on **6/9** (bold = wrong prediction).

#	Input	Gold	Qwen3-8B	Llama-3.1-8B
1	This umbrella kept me perfectly dry for the first ten seconds of the storm.	Neg	Pos	Neg
2	The battery lasts just long enough to get from my bedroom to the kitchen.	Neg	Pos	Neg
3	Customer support got back to me in only three hours, which was faster than my last ticket.	Neg	Pos	Pos
4	The phone survived a drop from pocket height, as long as I was sitting down at the time.	Neg	Pos	Neu
5	After only two visits to the mechanic this month, the car is running perfectly again.	Neg	Pos	Pos
6	The noise-cancelling headphones reduced the drilling noise next door to a very manageable level.	Neg	Pos	Pos
7	I have now successfully assembled three of the seven required pieces.	Neg	Pos	Neu
8	The waterproofing performed admirably in light drizzle, the only condition I had the chance to test.	Neg	Pos	Pos
9	I got to sample the entire menu, as each dish came back to the kitchen at least once.	Neg	Pos	Neg
Failures			8/9	6/9

Hypothesis Tests

Eight controlled inputs varying only the explicitness of negativity on one scenario (an umbrella). H1–H3 use explicit negative vocabulary (model should be correct); H4–H6 use pure understatement with no negative words (predicted failure); H7–H8 are positive controls.

#	Input	Gold	Qwen3-8B	Llama-3.1-8B
H1	This umbrella broke in the first minute of rain and I got completely soaked.	Neg	Neg	Neg
H2	This umbrella is useless. It fell apart immediately and left me drenched.	Neg	Neg	Neg
H3	Oh great, another umbrella that lasts all of ten seconds in the rain.	Neg	Neg	Neg
H4	This umbrella kept me dry for the first ten seconds of the storm.	Neg	Pos	Neg
H5	The battery lasts just long enough to reach the coffee machine from my desk.	Neg	Neg	Neg
H6	I used this umbrella once before going back to my old one.	Neg	Neg	Neu
H7	This umbrella is fantastic. It kept me completely dry through two hours of heavy rain.	Pos	Pos	Pos
H8	This umbrella handled a two-hour downpour without a single leak.	Pos	Pos	Pos

Both models correctly classify explicit negatives (H1–H3) and positive controls (H7–H8). Qwen3-8B fails on H4 as predicted; Llama produces **Neutral** on H6. The pattern confirms the hypothesis: *the more implicit the negativity, the more likely the model is to fail*.

Proposed Solution

Two-step prompting: first ask the model to paraphrase what the sentence implies about the product (“*In one sentence, describe what the following review implies about the product’s quality: [input]*”), then classify that generated text. The paraphrase will contain explicit evaluative language, making the classification straightforward. This requires no extra training, only one additional forward pass.